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**Textometric analysis of Swiss spatial
planning policies**

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Abstract

This study investigates how accessibility, motility, and time are represented in Swiss spatial planning discourse through a textometric approach.

It does so using a corpus made of 74 plans accross all cantons of switzerland spanning between 1987 to 2025. The results reveal a pronounced imbalance: accessibility dominates and is expressed in concrete, operational terms, while the other aspect of motility, skills, motivation, and cognitive appropriation is almost entirely absent. This absence shows that capability-based understandings of mobility remain outside the institutional vocabulary.

Time-related notions display a similar asymmetry. Objective and measurable aspects of time are frequently mentioned, whereas perceived or subjective time appears only marginally. Concepts such as comfort are present but vaguely defined, suggesting that experiential dimensions of mobility are acknowledged but not yet integrated into planning practice.

The textometric method proved effective in exposing what institutional discourse emphasizes and what it leaves unspoken. Furthermore, this approach, while not perfect, reduce researcher bias by offering some quantitative result.

The project contributes both by offering an analysis that show a discursive bias toward technical framings of mobility and an open, dataset with a CC-BY license to support further research.

Overall, the findings underline the need for conceptual and institutional shifts to incorporate motility and experiential time into planning. Doing so would help move Swiss mobility policy toward mobility policies that better account for citizen capabilities and experiential aspects of travel, and thus help transition to a more sustainable mobility.

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1 Introduction

1.1 Background: Spatial Planning and Transport Policies in Switzerland

Spatial planning in Switzerland operates in a layered fashion. The federal Spatial Planning Act (LAT: Loi sur l'aménagement du territoire) [2] was introduced in 1979 by the Confederation and apply to each Cantons and Communes. It defines the overall framework and general objectives such as containing urban sprawl, protecting natural resources, and ensuring efficient land use, but delegates the implementation to the Cantons. Each Canton is responsible for elaborating a "cantonal master plan" (plan directeur cantonal) within the first eight years of the LAT's introduction and then revising it at least every ten years. The Confederation must review and approve each plan and its revisions.

These plans translate federal principles into territorial strategies that account for local geography and specific needs. Once approved by the Confederation, they become binding for public authorities but remain indicative for private actors, shaping how subsequent zoning and land-use decisions are made. The Cantons delegate the responsibility for implementing some of the measures to the Communes, with the main exception of cantonal infrastructure such as cantonal roads, which remain under cantonal responsibility.

The Communes achieve the goals either by adopting local regulations (règlement communal) or by directly implementing measures, for instance by creating new bike lanes. Some Communes also prepare a "local master plan" (plan d'affectation communal), which materializes these strategies by determining the precise allocation of land and the regulations governing its use.

This nested structure creates offer flexibility to adapt the general goal as close as possible to the local reality but it require a fair amount of effort to ensure the efficient communication between all the actors. Over time, revisions of the LAT, most notably in 2014 [6], shifted the discourse from expansion toward densification, sustainability, and the coordination of land use with transport systems. The vast majority of early plans exist only as paper archives available only trough physical consultation, limiting historical analysis, but comparative studies between cantonal documents remain meaningful. Each plan reveals how spatial and transport policies adapt a shared legal foundation to diverse territorial realities and political cultures.

1.2 Research Motivation: Linking Travel Behavior, Planning, and Text Analysis

Effective policy design requires a systemic perspective. Interventions that optimize without taking into account the system as a whole often if not always generate unintended consequences[13]. The problem is actually very similar to what arise in artificial intelligence when systems end up acting in unanticipated ways due to failure (among other factor) of the cost function to capture the entirety of the human intent with the only goal of minimize the said cost function [12].

This principle directly applies to transportation policy. One of its main goals is to reduce environmental harm by lowering greenhouse gas emissions, which are the primary cause of human-induced climate change. The transport sector plays a key role in this effort. In Europe, it is the only major sector where emissions have not decreased, and it remains the second-largest emitter after energy production [3]. Decarbonizing transport is therefore essential for mitigating climate change. Furthermore, greenhouse gases are not the only negative externality. Transport also generates tire-related microplastics, fine particles, noise pollution, accident risks, soil sealing, and barriers to animal movement. These impacts must be balanced with other objectives derived from the right to mobility [16], such as equity and inclusivity, economic efficiency, and safety. Ultimately, transport policy must navigate a tradeoff between protecting the environment and its inhabitants from harmful externalities and ensuring the human need for mobility. This is not a trivial task as both factor are complex with many interconnected variables.

A transition to active mobility and public transport offers the most equitable path forward. Private vehicles are highly inefficient in their use of urban space. They provide primary benefits to a minority of users. This often occurs at a significant cost to the broader community [11]. Prioritizing sustainable modes is a matter of fairness and justice [8].

However, the existence of sustainable infrastructure does not guarantee its use. Research, such as the work of J. Grandvillemin and Al (not published yet), demonstrates this clearly. Modal choice is not a purely rational decision. It is shaped by intangible factors like personal habits, preferences, and perceived access. The sociological concept of motility effectively captures this capacity for mobility [10]. Understanding these human factors is crucial for designing effective systems.

This leads to the core research question of this document. To what degree are these human dimensions of mobility integrated into Swiss spatial planning? This study investigates whether cantonal master plans consider the factors that truly shape travel behavior. It aims to determine if planning discourse addresses how people experience time and choose between mobility options. The ultimate goal is to assess if Swiss policy aligns physical infrastructure with the human elements of mobility and thus to offer a path to improve the future versions of said plans.

1.3 Research Questions

To what extent do Swiss territorial / spatial planning documents acknowledge the distinction between perceived time vs measured/travel-time objectively modeled, in relation to modal choice?

How strongly do Swiss planning documents incorporate human / behavioral / motility-based rationales, as opposed to purely economic / efficiency rationales, in shaping transport and spatial policy?

2 Conceptual Framework & Key Definitions

This research is grounded in the sociology of mobility, which shifts the analytical focus from the mere movement by taking into account the individual. It's important to define and clarify a few concept first. The first one is Motility and the second one is Accessibility. In the general knowledge, mobility only take into account the Accessibility aspect and not the capacity for movement nor the social practices that shape it. This framework is essential for understanding the gap between the provision of transport infrastructure and its actual use. Motility regroup the accessibility and the user's skill and motivation to use that accessibility and together theses three notion become mobility.

2.1 The Concept of Motility

Motility is defined as the capacity of an individual or a group to be mobile in social and geographic space [10]. It is not the act of moving itself, but the potential for movement. This potential is structured by three interdependent elements:

- **Access:** This refers to the available mobility tools and opportunities within a specific context. It includes the physical transport infrastructure (e.g., roads, cycle paths, rail networks), the available services (e.g., timetables, pricing schemes, owning a vehicle), and the legal and institutional frameworks (e.g., driving licenses, residency permits). Access is the objective component of the system.
- **Skills:** These are the individual competencies required to utilize access. They include not only physical skills (e.g., the ability to cycle or drive) but also cognitive skills (e.g., reading a network map, use a ticket machine) and organizational skills (e.g., time management, trip optimization).
- **Cognitive Appropriation:** This is the most subjective and crucial element. It encompasses how individuals perceive, evaluate, and assign meaning to the access and skills available to them. It includes personal preferences, habits, values (e.g., environmental consciousness), perceptions of safety, and the social meanings attached to different modes of transport (e.g., the status of a car). To give a practical example, someone might prefer taking the train due to being able to re-appropriate the travel time by socializing with friends or doing administrative tasks.

Motility is, therefore, the product of the interaction between an individual's skills, the cognitive appropriation and the individual access.

2.2 Accessibility and the Institutional Perspective on Mobility

Accessibility represents the degree to which places, services, and opportunities can be reached within a given spatial and temporal context. Within the motility framework, accessibility corresponds to the *access* dimension: the tangible opportunities for movement available to an individual. It determines what movements are possible in principle, while skills and cognitive appropriation determine which of these possibilities become actual mobility practices.

In everyday discourse and most planning documents, however, mobility tends to be reduced to this accessibility dimension. When policymakers or engineers discuss "improving mobility," they often mean improving accessibility: reducing travel times, increasing network coverage, or enhancing service frequency. This reduction overlooks the skills, perceptions, and motivation that mediate how people actually use mobility systems. As a result, there can be a gap between the provision of infrastructure and its effective use.

From an institutional perspective, accessibility is usually assessed through measurable indicators. These include physical and temporal metrics such as travel time, cost, and service frequency (“qualité de la desserte”), but also considerations (since quite recently) of inclusivity such as the accessibility of transport for people with disabilities, affordability, and multimodal connectivity. These factors together define the “objective” structure of mobility opportunities in a given territory.

Yet, while accessibility can describe how well-connected a place is in physical terms, it cannot alone account for how people interpret and appropriate those connections. For example, two neighborhoods might have identical public transport coverage, but their residents may display very different travel behaviors due to differences in skills, habits, or perceptions of comfort and safety.

In this sense, accessibility forms the structural foundation of motility. As a mathematician would describe it : necessary but not sufficient for mobility to occur. Understanding this distinction is key to developing human-centered mobility policies that engage with how people experience, value, and enact movement in their daily lives and thus maximize the effectiveness of the measure taken.

2.3 Synthesis: Implications for Spatial Planning

This conceptual framework provides the lens through which Swiss planning documents will be analyzed. The key question is whether these documents operate with a simplistic model of mobility as movement, focusing primarily on access (e.g., building infrastructure) and objective travel time. Or, do they engage with the more complex, sociological model of motility and viscosity, which would involve considering how policies affect skills and cognitive appropriation, and acknowledge the role of perceived time in shaping modal choice? The presence or absence of this more nuanced vocabulary and rationale will be a key indicator of the human-centricity of planning policy.

3 State of the Art

Traditional conceptions of mobility were primarily based on metrics that fail to capture all the factors at play. The work of Geurs and van Wee [7] provides a comprehensive review of accessibility measures. They identified a gap in operational planning methods, which tend to be overly reductionist by using metrics such as travel speed or congestion level to facilitate the interpretation and communication of results. They showed that the quality and theoretical soundness of such evaluations could be greatly improved by taking into account behavioral and spatial dimensions, as well as the feedback loops that shape mobility, but their framework still does not account for the lived experience of travel.

This is where sociological approaches, from Sheller and Urry [19] and more notably Kaufmann's concept of motility [10] and Doherty's notion of viscosity [5], shine by providing a theoretical framework to explain the mismatch between predicted theoretical outcomes and observed results. Their work emphasizes how individuals experience travel and how their choices are shaped by these experiences. Jain and Lyons [9] further contribute to this through their research "The Gift of Travel Time," by challenging the idea that travel time is purely a waste and should therefore be minimized. They demonstrate how individuals can re-appropriate this time for meaningful activities, making perceived time fundamentally different from measured time. This distinction shows that the quality of the journey matters at least as much as its speed. Bühler [4] extends this perspective by noting the importance of habits and routines in modal choice, illustrating how decision-making is not purely rational but embedded in social and temporal practices.

Together, these works depict a more complete image of mobility through, first, the classical approach, which models accessibility as a measurable property, and second, the sociological approach, which presents mobility as a lived and experienced process.

The present study aims to assess how far these theoretical concepts have made their way into operational planning by analyzing a corpus of Swiss spatial policy documents, in order to determine whether planning discourse reflects a human-centered understanding of mobility or remains anchored in purely functional and physical-based reasoning. To do so, a total of forty words were selected from the vocabulary present in these publications and through exploration of the corpus and then mapped to a concept. With a few words per concept we can start to measure how much a concept is present in a text corpus as theorized in the textometric approach.

4 Methodology

4.1 Corpus construction

The corpus analyzed in this study consists of official cantonal master plans (*Richtpläne / plans directeurs cantonaux*) collected from the websites of all Swiss cantons. When digital versions were unavailable, cantonal offices were contacted directly. Documents available only in paper format were excluded due to logistical constraints. The corpus covers French and German documents; Italian documents from Canton Ticino were excluded as they cannot be meaningfully compared with other regions. A detailed list of documents is provided in Table 4.2, and all PDFs are available in the public repository [14].

In the early stages of the research, the corpus was analyzed using **TXM** [18], an open-source textometric platform designed for large-scale linguistic analysis. TXM was used for exploratory analysis, lemmatization, keyword frequency measurement, and correspondence analysis (AFC). However, due to limitation in filtering the corpus for the AFC and in order to improve the reproducibility and automation of the workflow, the pipeline was later reimplemented in Python. The final analysis therefore almost no longer depends on TXM with the exception of the lemmatization of the dataset, although TXM played an important role in shaping the initial methodological design.

4.2 Text extraction and preprocessing

Text was extracted from PDF documents using simple command-line tools and then annotated with metadata including canton, year, and language. All text was lemmatized using **TreeTagger** [17] and stored in XML format for structured processing.

The corpus was subsequently parsed in Python using `lxml` to build a structured token dataset containing lemma, part-of-speech (POS), and document-level metadata. POS statistics were computed to assess coverage and the proportion of unknown tokens, providing an overview of data quality.

4.3 dataset filtering, Token-level scoring, smoothing, and segmentation

Each token was assigned a score based on a curated whitelist mapping lemmas to numeric values, capturing their relevance to mobility and time concepts and mobility as a whole. This whitelist was made using TXM and the cocurrence tool to refine iteratively and find all the words that are related to our subject. To reduce noise and emphasize regions of concentrated conceptual relevance, scores were smoothed using a Gaussian kernel. High-scoring regions were then segmented based on a quantile threshold, producing subcorpora enriched in relevant lexical content. For each segment, the corresponding lemmas were exported in readable format to facilitate qualitative inspection.

4.4 Wordlist-based keyword filtering

Four custom wordlists were created for each language: one corresponding to terms related to **motility** and **accessibility** and another to terms referring to **perceived time** and **physical time**. These lists were manually assembled and iteratively refined based on corpus exploration and were made both in French and German. Both single-word and multi-word expressions were included. One could argue that this approach is influenced by the researcher, and would be correct for individual word. But since we have a word list, the averaging gets closer to the wanted concept. This is similar to psychometric test where one item in specific might not be able to predict meaningfully the presence of a disease like depression but ten items have a much stronger probability of measuring the latent concept.

4.5 Document-level counting and matrix construction

Within the extracted segments, occurrences of wordlist entries were counted per document. Counts were organized into matrices, with rows representing lemmas and columns representing documents. Separate matrices were constructed for each wordlist and aggregated into two main concept matrices: **Time** (perceived + physical) and **Mobility** (accessibility + motility). This was done for both French and German subcorpora.

4.6 Statistical analysis

Principal Component Analysis (PCA) was performed on each matrix using `scikit-learn`. PCA coordinates were normalized to a $[-1, 1]$ range for both documents and words. Visualizations display documents in blue and the top-loading lemmas in red. The explained variance of the first two components is reported to assess the dimensionality captured by the PCA. PCA was performed separately for French and German, as well as on merged bilingual matrices.

4.7 Reproducibility and automation

The entire workflow is implemented in Python and fully reproducible using open-source libraries: `lxml`, `pandas`, `numpy`, `matplotlib`, `scikit-learn`, `tqdm`, and `adjustText`. From XML parsing to KWIC generation, scoring, smoothing, segmentation, and PCA, all steps are automated in a single script (`analysis.py`). All intermediate and final outputs (segment texts, matrices, PCA plots, and text summaries) are saved in the `result/` folder to facilitate verification and replication.

Canton	Version	Year	Source	Language
AG	2	2025	Online	DE
	1	1996	†	DE
AI	4	2017	Online	DE
	3	2015	Online	DE
	2	2011	Online	DE
	1	2002	Online	DE
AR	5	2025	Online	DE
	4	2012	†	DE
	3	2008	†	DE
	2	2002	†	DE
	1	1987	†	DE
BE	10	2024	Online	FR
	10	2024	Online	DE
	8	2013	†	DE
	7	2012	†	DE
	6	2010	†	DE
	5	2008	†	DE
	4	2006	†	DE
	3	2004	†	DE
	2	2002	†	DE
	9	2016	†	FR
	8	2013	†	FR
	7	2012	†	FR
	6	2010	†	FR
	5	2008	†	FR
	4	2006	†	FR
	2	2002	†	FR
	1	1987	†	?
BL	2	2025	Online	DE
	1		†	DE
BS	7	2025	Online	DE
	6	2024	†	DE
	5	2019	†	DE
	4	2018	†	DE
	3	2012	†	DE
	2	2010	†	DE
	1	1986	†	DE
FR	3	2024	Online	FR
	2	2002	†	FR
	1	1987	†	FR
GE		2024	Online	FR
GL		2025	Online	DE
GR		2025	Online	DE
JU		2018	Online	FR

Canton	Version	Year	Source	Language
LU	4	2025	Online	DE
	3	2009	Online	DE
	2	1998	†	DE
	1	1986	†	DE
NE		2011	Online	FR
NW	4	2002	Online	DE
	3	2004	†	DE
	2	2000	†	DE
	1	1986	†	DE
OW	3	2019	Online	DE
	2	2006	†	DE
	1	1987	†	DE
SG		2017	Online	DE
SH	3	2021	Online	DE
	2	2004	†	DE
	1	198X	†	DE
SZ	6	2022	Online	DE
	5	2012	†	DE
	4	2010	†	DE
	3	2008	†	DE
	2	2004	†	DE
	1	1986	†	DE
SO	3	2023	Online	DE
	2	2000	†	DE
	1	1984	†	DE
TI	2	2009	Online	IT
	1	1990	†	IT
TG		2018	Online	DE
UR		2023	Online	DE
VD	8	2022	Online	FR
	7	2018	Online	FR
	6	2016	Online	FR
	5	2013	†	FR
	4	2012	†	FR
	3	2011	†	FR
	2	2008	†	FR
	1	198X	†	FR
VS		2019	Online	FR
ZG	3	2025	Online	DE
	2	2004	Online	DE
	1	1987	Online	DE
ZH		2020	Online	DE

Table 4.2: Cantonal master plans

Legend:

† Only available in paper format (physical consultation) and thus excluded from this study

‡ Sent upon request by the cantonal service.

* Did not answer to the email or the document whereabouts are unknown.

5 Results

5.1 Word Occurrences

In the first table 5.1 The results reveal a strong asymmetry between the two conceptual fields. Terms related to motility are almost absent from both corpora, while accessibility vocabulary is frequent and well represented. This suggests that Swiss spatial planning discourse focuses primarily on infrastructural and material dimensions of mobility. The human, cognitive, and motivational aspects that define motility are not expressed in the language of planning documents. Mobility appears as something provided through networks and infrastructure rather than as a capability that individuals develop and use. The conclusion is the same for French and German.

The same pattern is visible in the analysis of time-related terms in table 5.2. References to physical or objective time largely outweigh those to perceived or subjective time. Travel time is described as a variable to be reduced, optimized, or measured. Few terms evoke comfort or satisfaction. This indicates that time is approached mainly as an operational parameter rather than a human experience. But the contrast is not as dramatic as for motility between the two perspective.

Taken together, these findings show that planning discourse remains centered on measurable, supply-side factors. The subjective dimensions of mobility, including skills, motivation, and perception, are weakly represented not to say absent. The language used in official documents thus reflects and reinforces a narrow view of mobility, oriented toward infrastructure and performance rather than individual capacities and experiences.

Table 5.1: Frequency of mobility-related terms in French and German corpora (aligned by conceptual field: Motility vs Accessibility)

Category	FR term	FR hits	DE term	DE hits	Total
Motility					
	aptitude mobilité	3	Bewegungsfähigkeit	3	6
	capital mobilité	0	Mobilitätskapital	0	0
	habitus mobilité	0	Mobilitätshabitus	0	0
	autonomie déplacement	0	Bewegungsautonomie	0	0
	apprentissage mobilité	0	Mobilitätslernen	0	0
	stratégie déplacement	1	Mobilitätsstrategie	14	15
	compétence	159	Orientierungskompetenz	0	159
	représentation espace	0	Raumvorstellung	0	0
	motivation mobilité	0	Mobilitätsmotivation	0	0
	(valeur) symbolique mobilité	0	Symbolisch Mobilität	0	0
Subtotal (Motility)		163		17	180
Accessibility					
	accessibilité	284	Erreichbarkeit	207	491
	qualité desserte	159	Versorgungsqualität	3	162
	intégration transport	3	Verkehrsintegration	0	3
	coût	304	Kosten	321	625
	disponibilité transport	1	Verfügbarkeit verkehrsmittel	1	2
	infrastructure	1109	Infrastruktur	738	1847
	centralité spatial	0	räumlich zentralität	0	0
	inégalité accès	0	erreichbarkeitsgerechtigkeit	0	0
	budget mobilité	0	mobilitätsbudget	0	0
	réseau transport	331	verkehrsnetz	100	431
Subtotal (Accessibility)		2187		1370	3557

Table 5.2: Frequency of time-related terms in French and German corpora (aligned by concept)

Category	FR term	FR hits	DE term	DE hits	Total
Physical / Objective Time					
	Minute	132	Minute	207	339
	Heure	192	Stunden	0	192
	Durée	48	Dauer	2	50
	Vitesse	158	Geschwindigkeit	17	175
	Rapidité	17	Schnelligkeit	0	17
	Ponctualité	7	Pünktlichkeit	7	14
	Retard	4	Verspätung	1	5
	Kilomètre	180	Kilometer	28	208
	Ralentissement	3	Verlangsamung	3	6
	Cadence	160	Taktfrequenz	0	160
Subtotal (Physical)		901		265	1166
Perceived / Subjective Time					
	Attente	28	Wartezeit	3	31
	Stress	0	Fahrstress	0	0
	Confort	15	Komfort	0	15
	Expérience	20	Erfahrung	20	40
	Ressentir / Percevoir	5	Wahrnehmung	5	10
	agréable	10	angenehm	2	12
	Satisfaction usager	0	Fahrgastzufriedenheit	0	0
	Temps Réappropriable	0	Zeitnutzung	0	0
	temps valorisé	0	bewertete Zeit	0	0
	Temps vécu	0	gelebte Zeit	0	0
	Bien-être (mobilité)	0	Wohlfühlmobilität	0	0
Subtotal (Subjective)		78		30	108

5.2 Analysis of PCA Results related to time

As shown in Figure 5.1, the French corpus reveals a marked separation between objective and subjective temporal notions. Terms associated with measurable, operational time such as *vitesse*, *minute* and *kilomètre* are strongly loaded on the first principal component, highlighting their dominant role in structuring the discourse. In contrast, experiential notions like "stress", "confort", "attente" or "ressenti" cluster closely together and are much less present as seen in table 5.2. The cantonal points occupy a dense region in the lower-left quadrant, suggesting a relatively homogeneous and technically oriented use of time-related vocabulary across French-speaking institutional texts, with the exception of Bern, which leans more toward "cadence", "durée" and "vitesse" instead of "heure", "minute" or "kilomètre". The author theorizes that this may be due to a translation from German for the Bern plan, which is the only one available in both languages.

Figure 5.2 shows that the German corpus follows the same pattern, but with an even stronger emphasis on objective temporal measures. Terms such as "Geschwindigkeit" and "Kilometer" form the major axis of variation and clearly structure the first component. Subjective notions like "Erfahrung" or "angenehm" rarely appear and cluster together, suggesting that they contribute weakly to the overall variance. The clustering of cantonal abbreviations in a compressed zone indicates limited lexical differentiation among German-speaking institutional texts, reinforcing the observation that technical and operational understandings of time dominate the discourse, while experiential dimensions remain secondary with the main fracture being the usage of time versus distance.

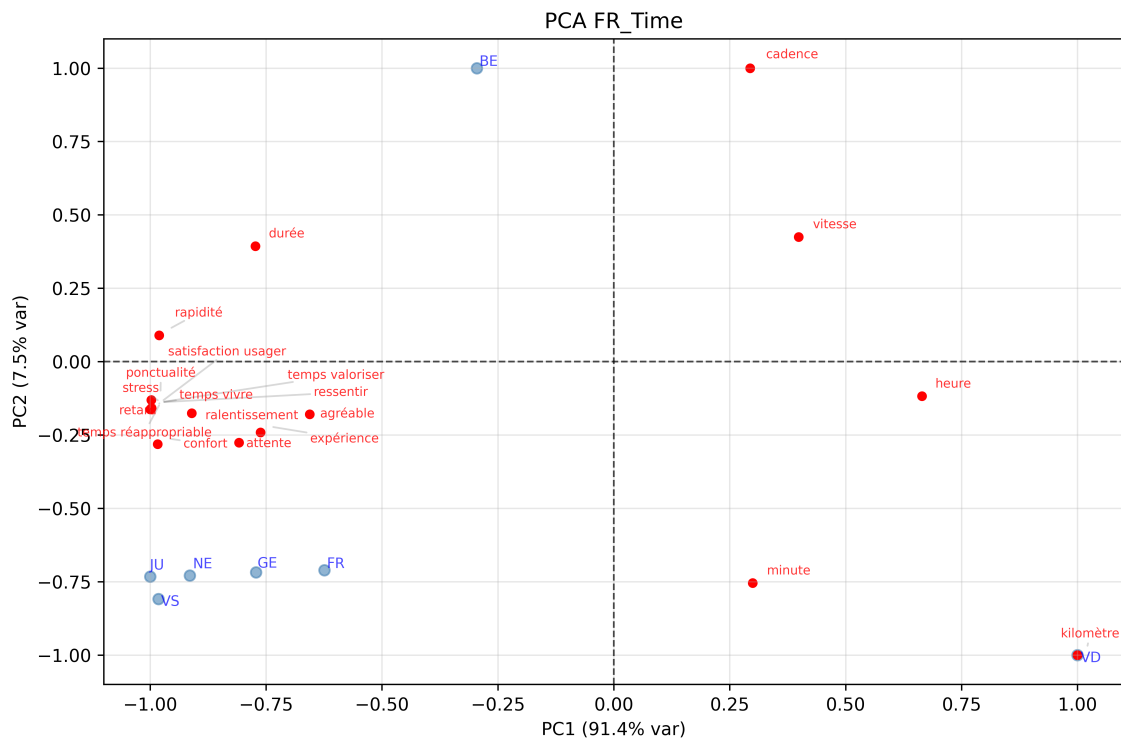


Figure 5.1: The French PCA highlights the dominance of quantifiable temporal notions, while subjective or experiential terms remain marginal in their contribution to variance.

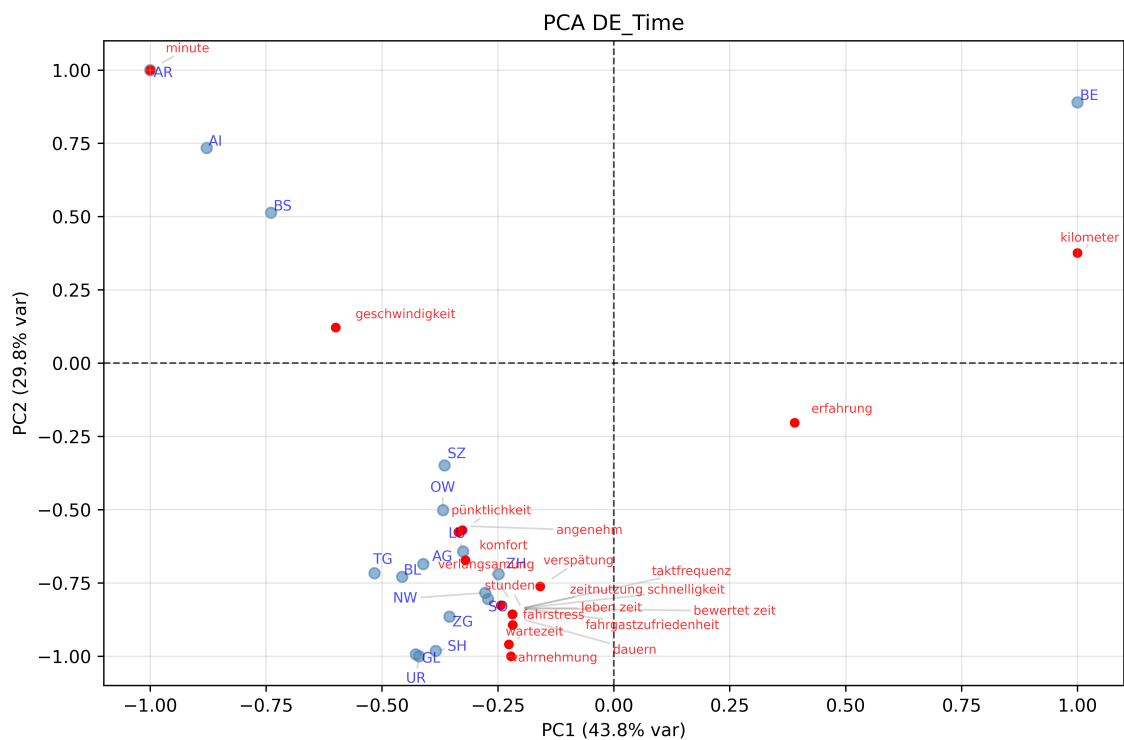


Figure 5.2: The German PCA confirms the strong prevalence of measurable time markers, with experiential notions appearing peripheral and weakly structuring the corpus.

5.3 Analysis of PCA Results related to mobility

Figure 5.3 shows a clear opposition between highly operational mobility notions and more capability-oriented terms. The first principal component captures the strong weight of "infrastructure", which dominates the positive side of the axis and forms the main structuring force in the corpus along with "coût", "accessibilité", "réseau transport" and "qualité desserte". Experiential or capability-related notions like "motivation mobilité", "apprentissage mobilité" or "habitus mobilité" cluster on the left very tightly together, indicating that they co-vary as a group but contribute only weakly to structuring differences between documents. The cantonal documents form a dense cluster near the negative side of the horizontal axis, except for Vaud and Bern, which are isolated at the positive extreme, similarly to the time PCA. This position suggests that VD relies more heavily on infrastructure and technical vocabulary than other French-speaking cantons, which use capability-based terms even more rarely than the others.

As shown in Figure 5.4, the German corpus displays a similar structure, with "infrastruktur" and "kosten" anchoring opposite sides of the second principal component. Capability-related terms such as "mobilitätskompetenz", "bewegungsautonomie" or "mobilitätshabitus" cluster near the centre and form a dense semantic block. Their central position indicates that they vary little across the documents and are very underrepresented per table 5.1; they are not the primary drivers of separation. The cantons again cluster tightly on the negative side of the horizontal axis, with BS, BL and BE positioned at the two extremes of the first two components. These outliers reflect a stronger emphasis on specific operational themes in these cantons, while the rest of the German-speaking corpus remains lexically homogeneous. Overall, measurable and infrastructural notions structure the corpus far more strongly than capability-related vocabulary.

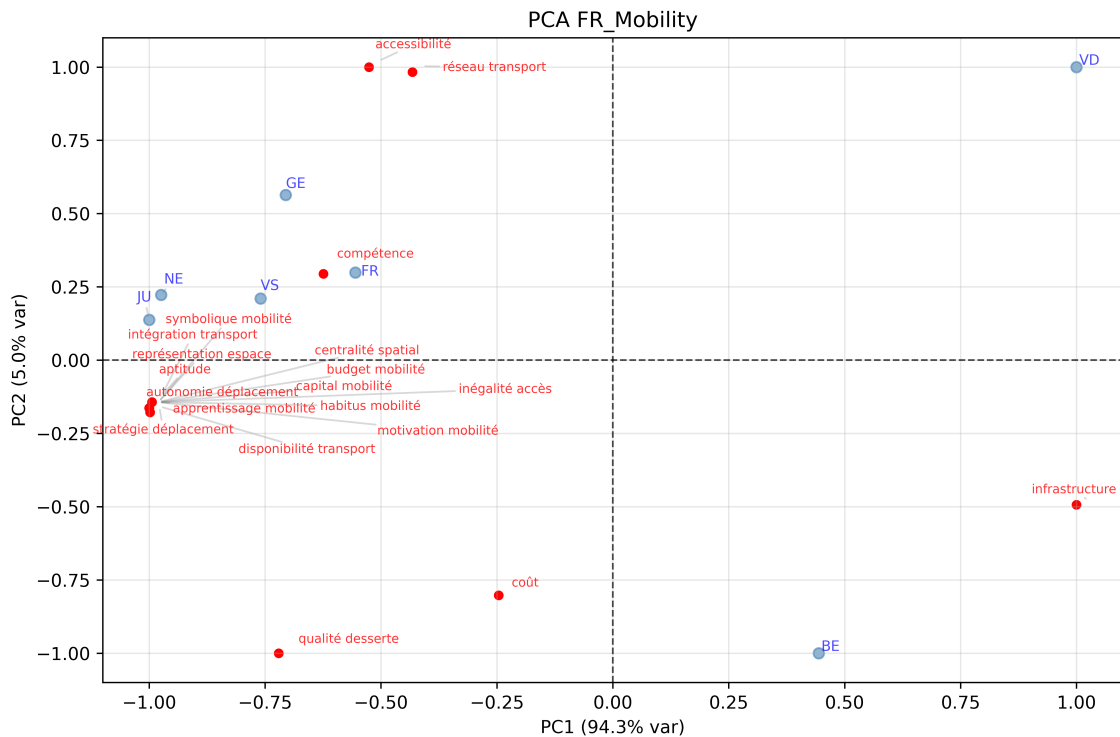


Figure 5.3: The French PCA reveals a strong axis contrasting infrastructure-focused terms with capability-oriented mobility notions, with VD standing apart from the other cantons.

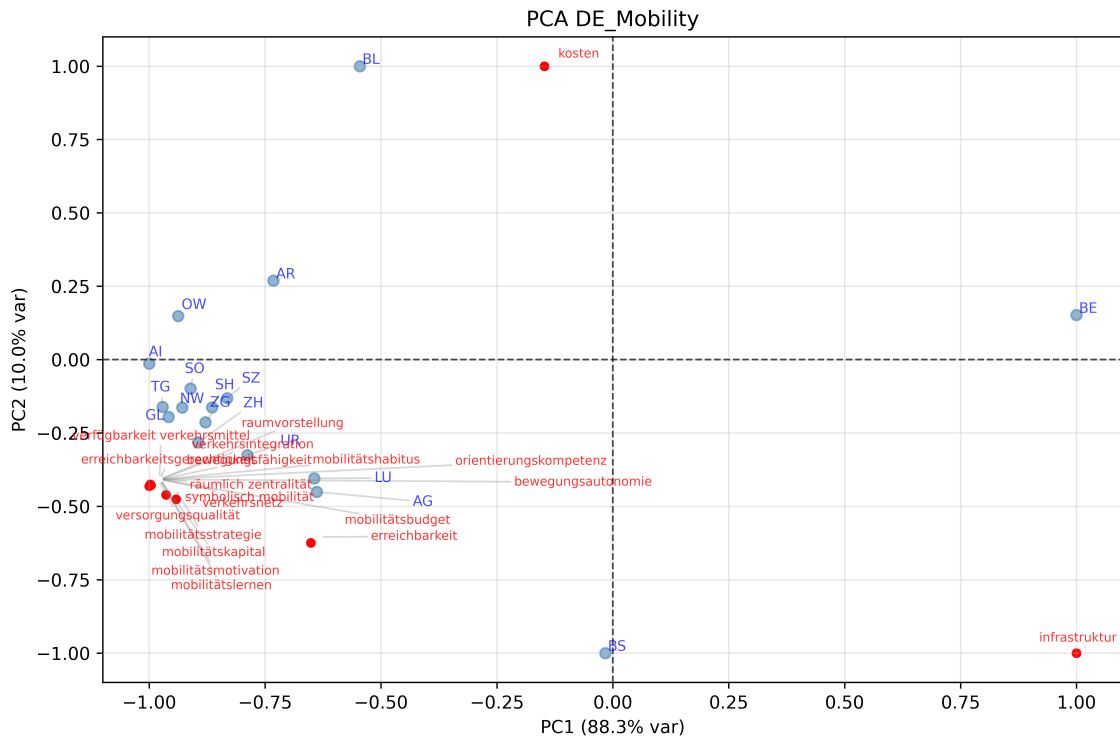


Figure 5.4: The German PCA highlights the dominance of operational and infrastructural mobility terms, while capability-related notions remain central and weakly discriminant.

5.4 Synthesis

Combining the AFC results with the word occurrence analysis reveals a coherent narrative: Swiss spatial planning discourse is heavily skewed toward measurable, supply-oriented dimensions of mobility. Accessibility and physical time terms dominate both corpora, while motility and subjective time concepts are almost absent. This is consistent across French and German texts.

Motility is conceptually underrepresented, indicating that planners primarily consider mobility as a function of infrastructure, services, and network performance rather than individual capabilities, skills or motivation. Time is similarly treated as a quantifiable variable, with minimal attention to comfort, perception or human experience.

The combined word frequency and AFC results reinforce the conclusion that Swiss planning documents operationalize mobility and time in terms of provision and efficiency. Human-centered aspects, including cognitive, motivational and experiential dimensions, remain peripheral, reflecting a narrow and infrastructure-focused view of mobility.

6 Discussion

6.1 Limitations of the Keyword Approach

The first limitation of this study concerns the use of keyword frequency as a proxy for conceptual presence. The textometric approach offers a reproducible way to measure linguistic patterns but counting words alone cannot ensure that they refer to the same idea across contexts. A high frequency of a term does not always mean that the underlying concept is strongly represented.

Future work should attempt at addressing this issue. This would strengthen the link between observed data and theoretical concepts. More advanced computational methods, such as contextual embeddings or supervised models, could help by taking into account how meaning shifts across contexts.

6.2 Challenges in Modeling Motility

A second limitation relates to how motility is modeled. Currently, there is no shared quantitative model of motility. The framework proposed by Kaufmann and later authors remains mainly qualitative, with the exception of accessibility, focused on the interaction between access, skills, and cognitive appropriation. Turning these elements into measurable indicators is a challenge.

Even if a complete model existed, combining the different dimensions into one global indicator would be problematic. From a philosophical point of view, there is no neutral way to collapse complex human phenomena into a single number. Each weighting reflects implicit values. For that reason, a multi-criteria approach such as Pareto fronts may be viable. It identifies solutions that are not dominated by others without imposing a single hierarchy. But it can only help to weed out the solution that are equal or inferior on every metrics. Since most solutions are tradeoff, they tend to be better in one metrics at the expense of another which Pareto front cannot solve. The final choice then remains with human decision-makers, who must accept the responsibility and bias that come with judgment.

6.3 Implications for Swiss Spatial Planning

The results suggest that Swiss spatial planning discourse still gives more attention to accessibility in its physical and infrastructural sense than to the human factors of mobility. Cantonal master plans mostly focus on network coverage, travel time, and service quality. They rarely address the motivation or skills that allow people to use these systems effectively. This shows that planning practice remains centered on supply-side measures, while the demand-side, related to behavior and perception, is often neglected.

Integrating motility into planning would mean designing measures that develop not only access but also the ability and willingness to use sustainable transport. This could involve highlighting the personal benefits of public transport and active mobility, and creating conditions that make their use natural and rewarding in everyday life.

6.4 Addressing the under-representation of Motivation and Skills

The under-representation of the “motivation” and “skills” dimensions of motility can be partly mitigated through targeted interventions. Measures such as training programs for using public or active transport, and introductory mobility workshops can help build both competence and confidence and help shape new habits. free or reduced-cost access to public transport could also help by both helping the accessibility of public transport and the motivations of the users. Interventions during key life stages, for example in schools, could encourage lasting habits of sustainable travel.

Existing research in “mobility management” [1] [15] shows that soft incentive to limit car use exist but their effect is debated [20]. Kaufmann’s approach could help improve the effectiveness of such measure by addressing the reason why people find that car are a better tradeoff from their perspective, and thus offer more effective measure that feel less frustrating to the user.

Further work is needed to understand how these ideas could be applied systematically within Swiss planning. Operationalizing motility as a concrete planning objective remains an open challenge, both theoretical and practical.

7 Conclusion

This study examined how accessibility, motility, and time are represented in Swiss spatial planning discourse using a textometric approach. The analysis reveals a clear imbalance in planning language. Accessibility is widely present and discussed in concrete terms, whereas motility, which encompasses skills, motivation, and cognitive appropriation, is almost entirely absent. This absence indicates that capability-based dimensions of mobility are not yet part of the institutional vocabulary, making them difficult to plan for or evaluate.

Time is represented asymmetrically as well. References to physical and objective time appear frequently and are treated as operational variables to be measured or minimized. Perceived or subjective time appears to a lesser extent. Concepts such as comfort are mentioned but remain vague and underdeveloped. This suggests that planners have begun to acknowledge experiential aspects of travel, yet they do not articulate them in operational terms.

Methodologically, the textometric pipeline proved effective at revealing patterns of presence and absence in institutional language. It is valuable for mapping what planning discourse renders thinkable and actionable. At the same time, the approach is essentially descriptive. Combining it with contextual and semantic analyses would better recover meaning beyond raw counts, reducing reliance on manual interpretation and mitigating researcher bias.

The project's contributions are twofold. First, it documents a persistent discursive bias in official planning documents toward supply-side and technical framings of mobility. Second, it provides a reproducible dataset and analysis pipeline, released under a CC-BY license in the project repository [14], enabling further research or replication. Making this corpus freely available lowers the "cost" of entry and allows exploration of broader aspects of spatial planning beyond mobility alone.

Looking forward, bridging the gap between accessibility and motility requires both conceptual and institutional change. We need ways to measure motility without oversimplifying it, and planning should routinely collect data on experiential and behavioral dimensions without falling into's Goodhart's law. Only then can mobility policy move beyond networks to truly support people's capabilities and maybe let the author sleep a little easier knowing that Switzerland point a bit more toward a sustainable future.

8 Appendix

8.1 logbook

2025-08-27 kickoff meeting

2025-09-13 Mooc textometrie, recherche des sources de document

2025-09-15 Investigation de la structure entre les documents, Formation du Dataset

2025-09-25 Creation of this report, rough layout of the idea in chapter + meeting with supervisor

2025-10-02 Formation recherche Zotero

2025-10-05 Définition du champ l'exical en rapport aux questions de recherche, Analysis in TXM, searching for patterns

2025-10-15 Presentation mis semestre kaufmann + Jules

2025-10-16 Script python, run against word list

2025-10-16 Made motilité word list

2025-10-22 Made german list + analysis, sent email for previous version

2025-10-24 Writing

2025-10-26 Sorting PDF + AFC setup using TXM (failure)

2025-10-27 Writing, re-run analysis with new document

2025-10-28 finished new script to make clean AFC in python

2025-10-29 added one more document received by email, writing methodology

2025-11-07 Visio avec Thomas Bühler

2025-11-10 rework AFC + wordlist

2025-11-17 analysis, result writing

8.2 scripts

[Listing 1:](#) Extract text content from PDFs by language

```
mkdir -p ../TXT && for f in *.pdf; do lang="${f:0:2}";  
mkdir -p "../TXT/$lang";  
pdftotext "$f" "../TXT/$lang/${f%.pdf}.txt"; done
```

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